

Ema Lovrić, 17.6.2026.

Assignment 11: Network monitoring with Wireshark

<https://github.com/elovric13/eutech/tree/network-monitoring-with-wireshark-elovric13>

1. Install Wireshark on a workstation or VM with network access

Wireshark was installed on an Ubuntu VM. During installation, non-superuser packet capture was enabled, and the current user was added to the wireshark group to allow capturing without running as root.

The active network interface (ens33) was selected from the capture interface list and used for all captures in this report.

2. Capture network traffic for at least 10 minutes

Traffic was captured for over 10 minutes.

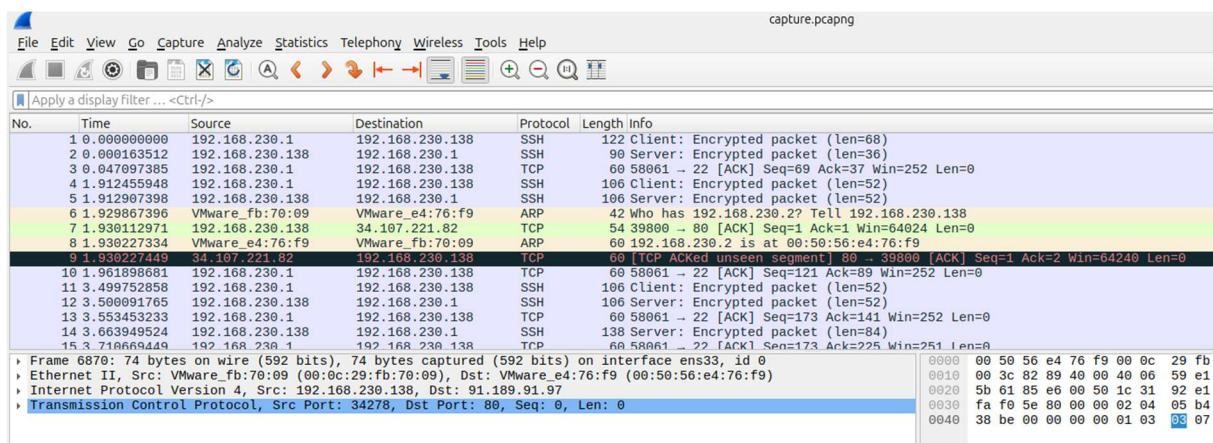
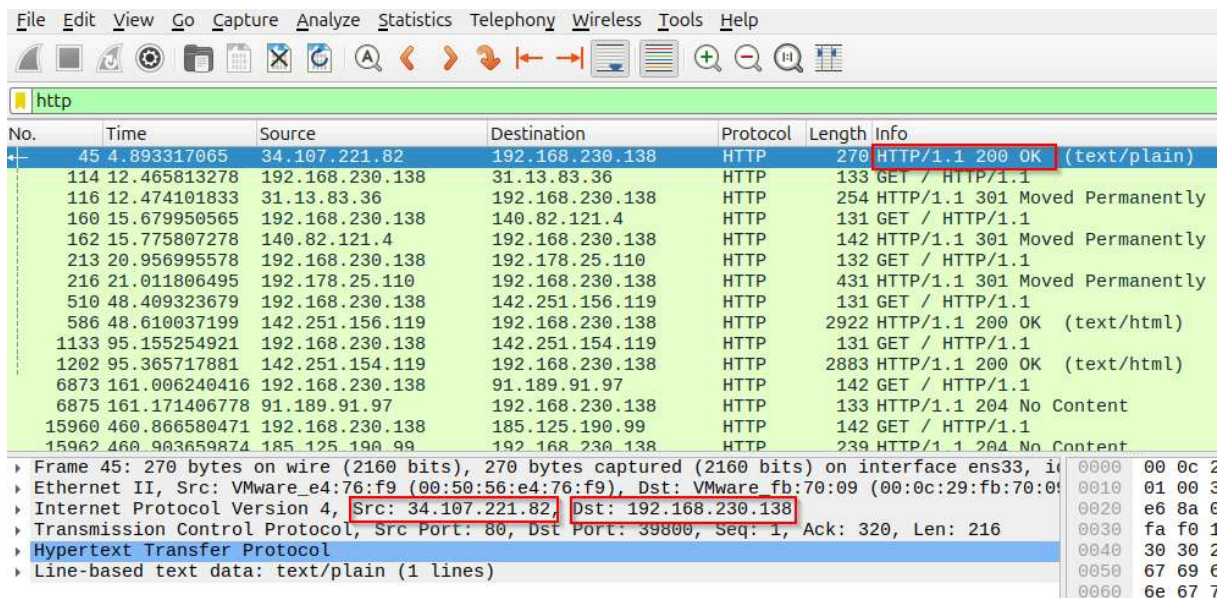


Figure 1: Wireshark capture

3. Filter and analyze packets related to HTTP/HTTPS, SSH, and DNS

- HTTP traffic

The http filter was applied to isolate HTTP requests and responses. The capture showed multiple GET requests from the local machine (192.168.230.138) to various destination IPs, with corresponding 200 OK, 301 Moved Permanently, and 204 No Content responses.



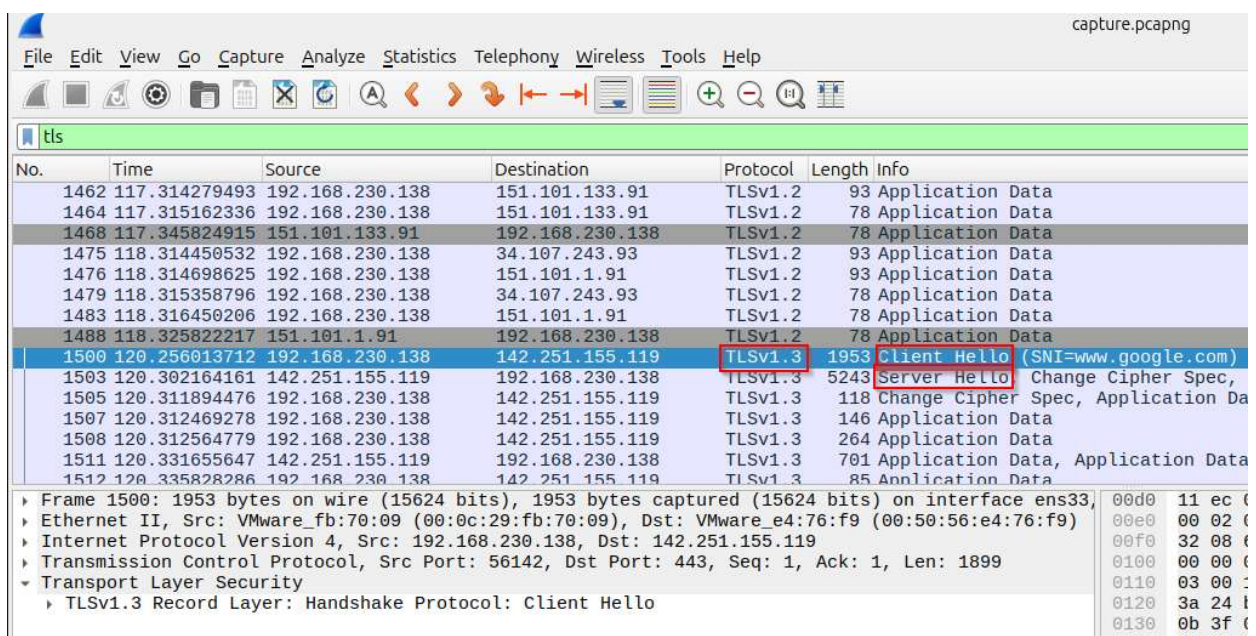
No.	Time	Source	Destination	Protocol	Length	Info
45	4.893317065	34.107.221.82	192.168.230.138	HTTP	270	HTTP/1.1 200 OK (text/plain)
114	12.465813278	192.168.230.138	31.13.83.36	HTTP	133	GET / HTTP/1.1
116	12.474101833	31.13.83.36	192.168.230.138	HTTP	254	HTTP/1.1 301 Moved Permanently
160	15.679950565	192.168.230.138	140.82.121.4	HTTP	131	GET / HTTP/1.1
162	15.775807278	140.82.121.4	192.168.230.138	HTTP	142	HTTP/1.1 301 Moved Permanently
213	20.956995578	192.168.230.138	192.178.25.110	HTTP	132	GET / HTTP/1.1
216	21.011806495	192.178.25.110	192.168.230.138	HTTP	431	HTTP/1.1 301 Moved Permanently
510	48.409323679	192.168.230.138	142.251.156.119	HTTP	131	GET / HTTP/1.1
586	48.610037199	142.251.156.119	192.168.230.138	HTTP	2922	HTTP/1.1 200 OK (text/html)
1133	95.155254921	192.168.230.138	142.251.154.119	HTTP	131	GET / HTTP/1.1
1202	95.365717881	142.251.154.119	192.168.230.138	HTTP	2883	HTTP/1.1 200 OK (text/html)
6873	161.006240416	192.168.230.138	91.189.91.97	HTTP	142	GET / HTTP/1.1
6875	161.171406778	91.189.91.97	192.168.230.138	HTTP	133	HTTP/1.1 204 No Content
15960	460.866580471	192.168.230.138	185.125.190.99	HTTP	142	GET / HTTP/1.1
15962	460.903659874	185.125.190.99	192.168.230.138	HTTP	239	HTTP/1.1 204 No Content

Frame 45: 270 bytes on wire (2160 bits), 270 bytes captured (2160 bits) on interface ens33, in frame 45
Ethernet II, Src: VMware_e4:76:f9 (00:50:56:e4:76:f9), Dst: VMware_fb:70:09 (00:0c:29:fb:70:09)
Internet Protocol Version 4, Src: 34.107.221.82, Dst: 192.168.230.138
Transmission Control Protocol, Src Port: 80, Dst Port: 39800, Seq: 1, Ack: 320, Len: 216
Hypertext Transfer Protocol
Line-based text data: text/plain (1 lines)

Figure 2: HTTP filter

- HTTPS traffic

The tls filter was applied to isolate encrypted HTTPS traffic. The capture clearly showed a TLS 1.3 handshake, including a Client Hello packet with the SNI (Server Name Indication) field revealing the requested domain, followed by a Server Hello response from the destination server.



No.	Time	Source	Destination	Protocol	Length	Info
1462	117.314279493	192.168.230.138	151.101.133.91	TLSv1.2	93	Application Data
1464	117.315162336	192.168.230.138	151.101.133.91	TLSv1.2	78	Application Data
1468	117.345824915	151.101.133.91	192.168.230.138	TLSv1.2	78	Application Data
1475	118.314450532	192.168.230.138	34.107.243.93	TLSv1.2	93	Application Data
1476	118.314698625	192.168.230.138	151.101.1.91	TLSv1.2	93	Application Data
1479	118.315358796	192.168.230.138	34.107.243.93	TLSv1.2	78	Application Data
1483	118.316450206	192.168.230.138	151.101.1.91	TLSv1.2	78	Application Data
1488	118.325822217	151.101.1.91	192.168.230.138	TLSv1.2	78	Application Data
1500	120.256013712	192.168.230.138	142.251.155.119	TLSv1.3	1953	Client Hello (SNI=www.google.com)
1503	120.302164161	142.251.155.119	192.168.230.138	TLSv1.3	5243	Server Hello
1505	120.311894476	192.168.230.138	142.251.155.119	TLSv1.3	118	Change Cipher Spec, Application Data
1507	120.312469278	192.168.230.138	142.251.155.119	TLSv1.3	146	Application Data
1508	120.312564779	192.168.230.138	142.251.155.119	TLSv1.3	264	Application Data
1511	120.331655647	142.251.155.119	192.168.230.138	TLSv1.3	701	Application Data, Application Data
1512	120.335828286	192.168.230.138	142.251.155.119	TLSv1.3	85	Application Data

Frame 1500: 1953 bytes on wire (15624 bits), 1953 bytes captured (15624 bits) on interface ens33, in frame 1500
Ethernet II, Src: VMware_fb:70:09 (00:0c:29:fb:70:09), Dst: VMware_e4:76:f9 (00:50:56:e4:76:f9)
Internet Protocol Version 4, Src: 192.168.230.138, Dst: 142.251.155.119
Transmission Control Protocol, Src Port: 56142, Dst Port: 443, Seq: 1, Ack: 1, Len: 1899
Transport Layer Security
TLSv1.3 Record Layer: Handshake Protocol: Client Hello

Figure 3: TLS handshake

- DNS traffic

The dns filter was applied to isolate domain name resolution traffic. The capture showed standard DNS queries and responses, confirming domain names being resolved to their corresponding IP addresses by the local DNS resolver (192.168.230.2).

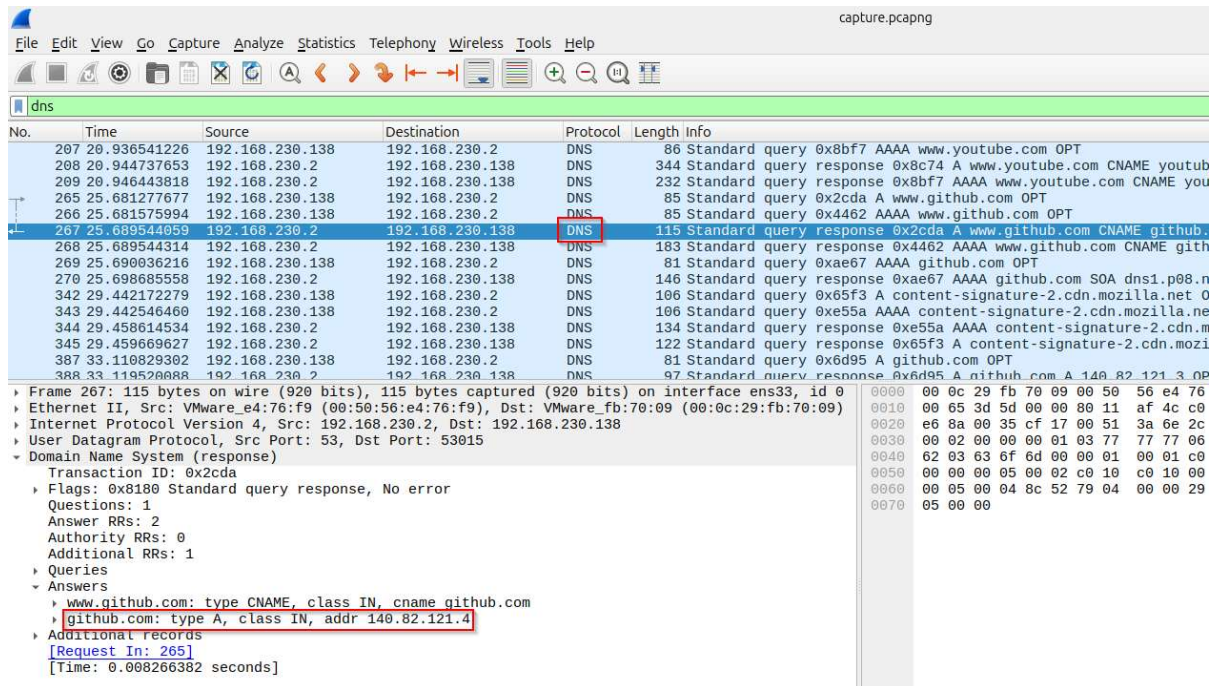


Figure 4: DNS query response

- SSH traffic

The ssh filter was applied to isolate SSH traffic from a local SSH connection to localhost. As expected with SSH, no readable content was visible since all packets were encrypted, shown as 'Encrypted Packet' in both the client-to-server and server-to-client directions.

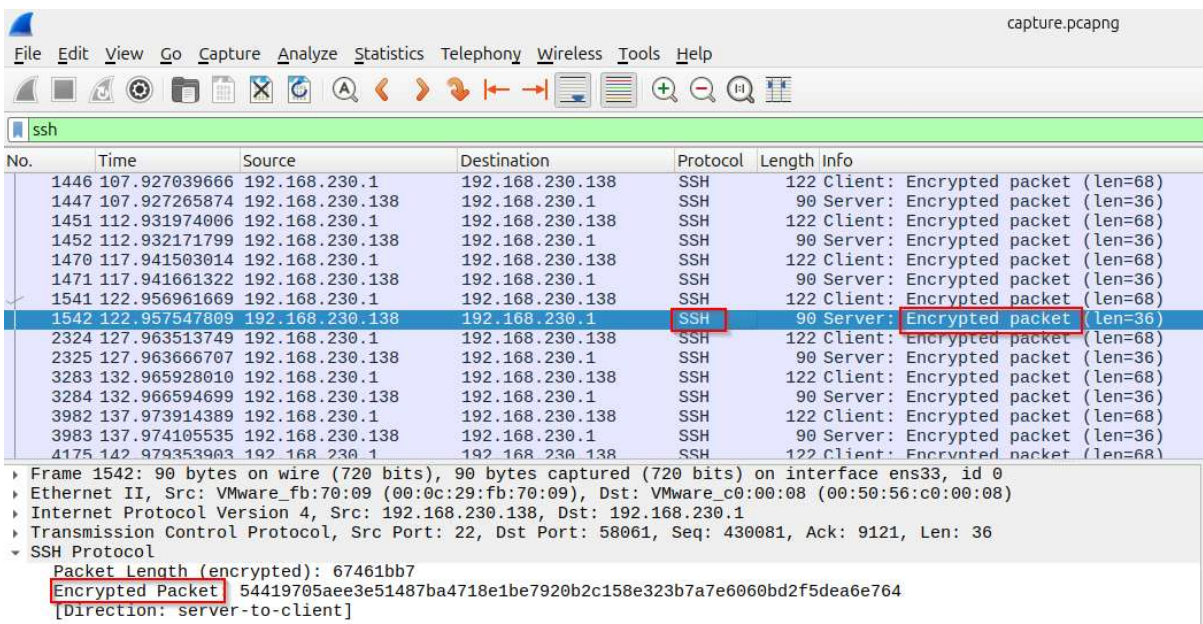


Figure 5: SSH traffic

4. Identify IP addresses, protocols, and possible suspicious access attempts

Protocol	Local IP	Remote IP	Purpose
HTTP	192.168.230.138	34.107.221.82	Plaintext web request
TLS/HTTPS	192.168.230.138	142.251.155.119	Encrypted web request (google.com)
DNS	192.168.230.138	192.168.230.2	Domain name resolution
SSH	192.168.230.138	192.168.230.1	Encrypted remote shell session

- Checking for suspicious access attempts

The filter `tcp.flags.syn == 1 && tcp.flags.ack == 0` was used to isolate new outbound connection attempts. All SYN packets were directed to standard ports (80, 443) across expected destination IPs, with no repeated connections to a single IP and no unusual ports. No suspicious activity was identified.

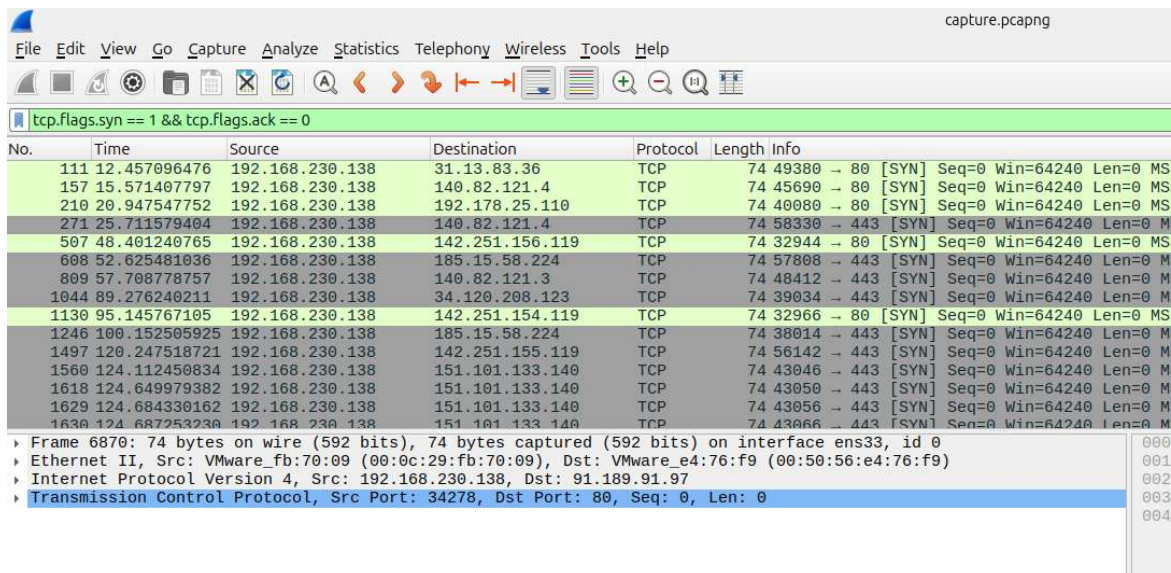


Figure 6: SYN filter

This task demonstrated the use of Wireshark to capture and analyze live network traffic. By applying protocol-specific filters, it was possible to clearly isolate and examine HTTP, HTTPS, DNS, and SSH traffic, identify the IP addresses involved in each type of communication, and verify that no suspicious connection patterns were present in the captured traffic.